8 **(a)** A beam of unpolarised, monochromatic light is passed through three polarisers. The intensities of the beam emerging from each polariser are I_0, I_1 and I_2 respectively.

The polarising axis of each polariser is shown by an arrow and held at different angles to the vertical axis, as shown in Fig. 8.1.

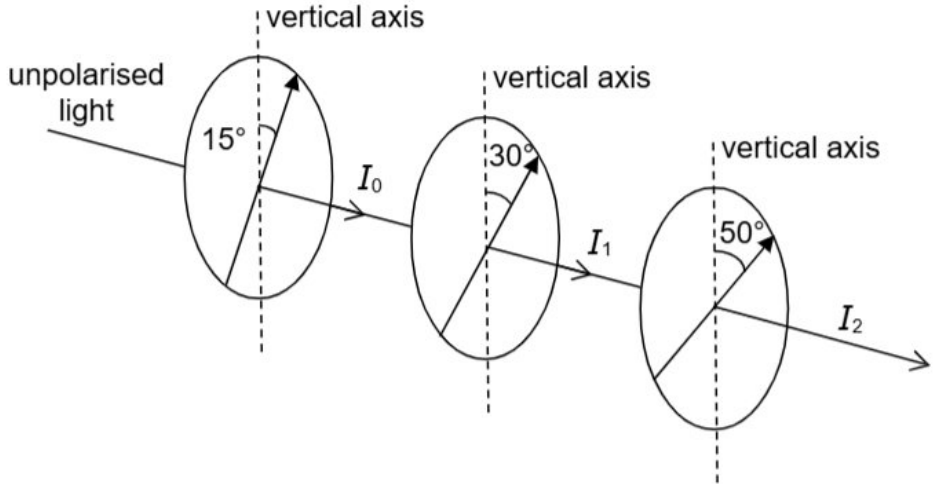


Fig. 8.1

(i) Explain what *polarisation* is.

.....

.....

..... [1]

(ii) A student believes that sound waves can be polarised.

State and explain whether this student is right.

.....

.....

..... [2]

(iii) State the intensity of the unpolarised light entering the first polariser in terms of I_0.

intensity = ... [1]

(iv) Express I_2 in terms of I_0.

I_2 = ... [2]

(b) The polarised monochromatic light of intensity I_2 is incident normally on a diffraction grating as shown in Fig. 8.2. Bright spots can be observed on the screen.

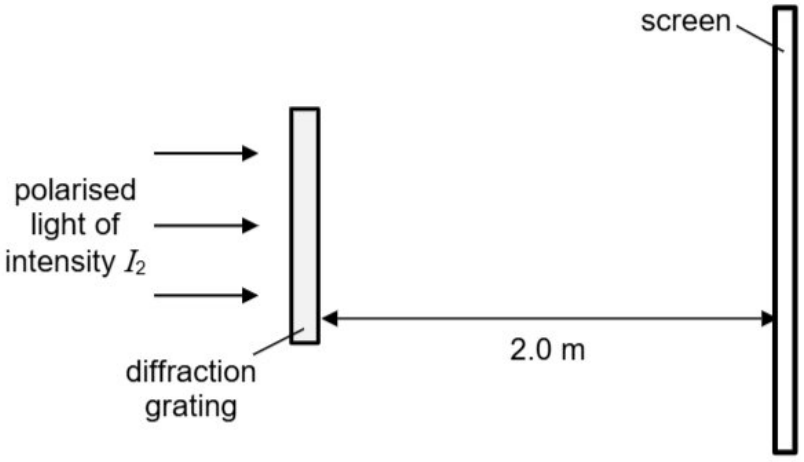


Fig. 8.2

The diffraction grating has 4.5×10^5 lines per metre. The first order bright spots are 0.60 m away from the central bright spot.

(i) Show that the wavelength of the light that passes through the diffraction grating is 640 nm.

[2]

(ii) Determine the maximum number of bright spots that can observed on the screen.

maximum no. of bright spots = ... [3]

(iii) The monochromatic light is now replaced by a single white light source.

State and explain what will be observed on the screen.

.....

.....

.....

.....

.....

..... [3]

(c) The same monochromatic light from **(a)** and **(b)** is now incident normally on a double slit, as shown in Fig. 8.3. The variation of intensity with distance along the screen is shown in Fig. 8.4.

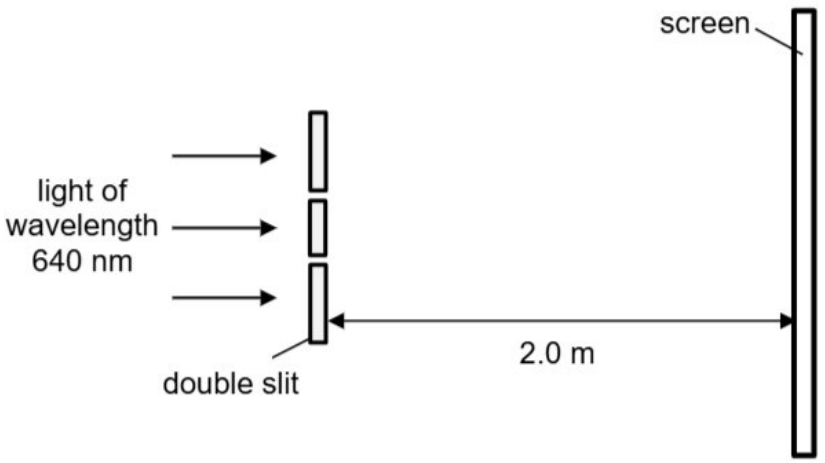


Fig. 8.3

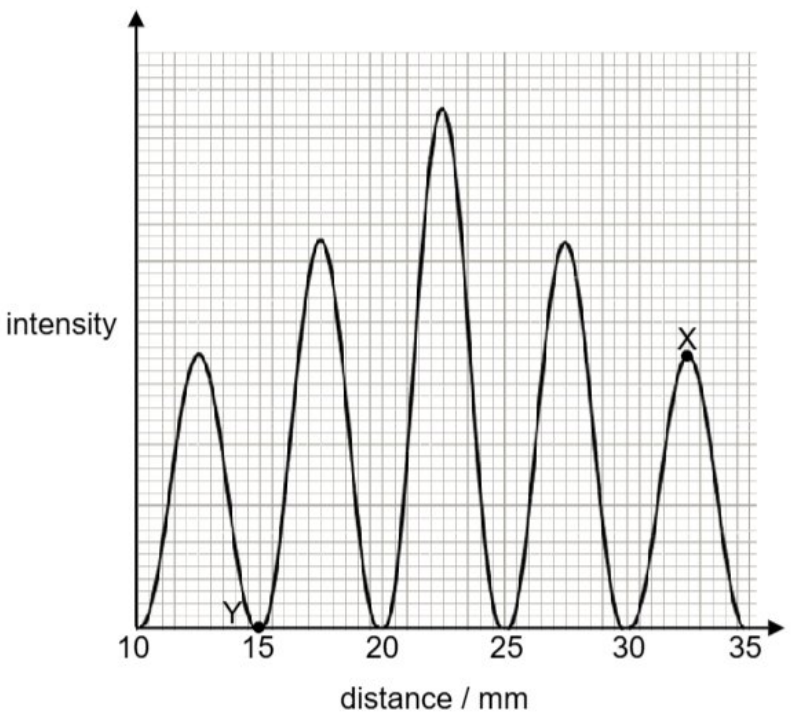


Fig. 8.4

(i) Determine the separation between the two slits.

separation = ... m [2]

(ii) Determine the path difference between the waves from the slits when the waves meet to produce the intensity at point X in Fig. 8.4.

path difference = ... m [1]

(iii) One of the slits is now covered with a thin piece of tracing paper.

(A) State and explain how the intensity at point Y will change.

.....

.....

.....

..... [2]

(B) State how the distance between adjacent bright fringes will change.

.....

..... [1]

[Total: 20]